APPROVED: 19 February 2020
doi:10.2903/sp.efsa.2020.EN-1810

FoodEx2 maintenance 2019

European Food Safety Authority (EFSA),
Marina Nikolic and Sofia Ioannidou
EFSA Evidence Management Unit

Abstract

FoodEx2 is a comprehensive food classification and description system developed by FESA that needs to be actively maintained, remain up to date as well as flexible to accommodate data collections in different domains. To this end regular maintenance of the system is needed. A first maintenance was carried out in 2015 and a second during the years 2016-2018 in order to evaluate further comments and proposals provided to EFSA by users and stakeholders. This technical report describes the outcome of the third maintenance process done during 2019 which entailed addition of new terms, deprecation of terms and amendments to existing terms including enrichment of implicit facets.

Key words: food classification, food description, FoodEx2

Requestor: EFSA

Question number: EFSA-Q-2020-00085

Correspondence: data.collection@efsa.europa.eu

Acknowledgements: EFSA wishes to thank the EFSA staff members Montserrat Anguita Freixa, Inmaculada Aznar Asensio J, Daniela Brocca, Davide Gibin, Paola Medina, Giuseppe Triacchini, Anca Violeta Stoicescu for the support provided to this report and all the users and stakeholder of FoodEx2 who provided suggestions for the maintenance process.

Suggested citation: EFSA (European Food Safety Authority), Nikolic M and Ioannidou S, 2020. FoodEx2 maintenance 2019. EFSA supporting publication 2020:EN-1810. 20 pp. doi:10.2903/sp.efsa.2020.EN-1810

Summary

FoodEx2 is a standardised food classification and description system developed by EFSA to improve the comparability and exchange of data coming from different food safety domains within the remit of EFSA. In order to meet this requirement, it needs to be actively maintained, remain up to date as well as flexible to accommodate data collections in different domains. To this end maintenance of the system has taken place since April 2015 when the major revision (FoodEx2 revision 2) was performed.

A maintenance technical group composed of EFSA staff with expertise in different domains (food consumption, chemical contaminants, pesticides residues, etc.) was set-up to evaluate proposals and requests provided by users and stakeholders and take decisions regarding their implementation. The first maintenance was carried out in 2015 and a second during the years 2016-2018.

During the third maintenance process performed in 2019 the following changes have been implemented: the scope of some existing terms was better specified or expanded by changing the name, scientific name, adding alias or other implicit attributes; two terms were dismissed i.e. set as not reportable in some of the hierarchies; two were deprecated due to duplication and another one was deprecated due to alignment with the Commission Regulation (EU) 2018/1497; 587 new terms were added in different hierarchies and facets, most of which related to bird species under the Avian Influenza (AI) surveillance programme; the applicability of some terms was extended to further hierarchies and/or facets and the position or type of some terms in the hierarchy tree was changed since a more appropriate parent-child term relationship was found; implicit facets were updated for some terms for their more accurate description and understanding by users and the structure and content of some FoodEx2 sections were amended.

In addition, a major revision was undertaken in the Feed hierarchy according to the latest version of the Catalogue of Feed materials (Commission Regulation (EU) 2017/1017 that amended Commission Regulation (EU) No 68/2013). In order to keep track of the assigned number of terms from the Catalogue of Feed Materials, new implicit attribute - EUFeedReg was created and added to all terms that originated from this source. Moreover, Feed hierarchy was reviewed in terms of the OECD GD classification (Organisation for Economic Co-operation and Development Guidance Document) and International Feed Nomenclature (IFN) codes were added as a new implicit attribute to all terms were the direct mapping with FoodEx2 was found. The facet F23 – Target-consumer was also amended to include more detailed classification of consumers (animal species) in the feed area.

The present report describes the outcome of the third maintenance process done in 2019 in order to keep track of the changes in the system within three minor releases published in 2019 (MTX 10.1, MTX 10.2 and MTX 10.3) and the major release published in January 2020 (MTX 11.0).

Table of contents

1. Introduction

1.1. Background and Terms of Reference as provided by the requestor

In 2011, EFSA released a comprehensive food classification and description system named FoodEx2 (revision 1) (EFSA, 2011a, b). This system was developed with the aim of harmonising the description of the food matrices within data collections in different food safety domains. Following the publication, a testing phase was carried out in order to highlight strengths and weaknesses of the classification and to identify possible issues and needs for refinement. Based on the outcome of the testing phase, EFSA revised the system and in 2015 the second revision of the food classification and description system (FoodEx2 revision 2) was published (EFSA, 2015).

According to the guidance documents on FoodEx2 (EFSA, 2011a, 2015), internal and external users should provide to EFSA proposals for implementation and maintenance in order to refine the system and an annual EFSA maintenance technical group should be set up in order to evaluate the suggestions and decide on their implementation. In fact, innovation in the food market and gained experience by using the system led to the need of keeping the catalogue updated. Following the revision 2 of the system, the FoodEx2 catalogue was revised several times based on EFSA needs and evaluation of the data providers and stakeholders' suggestions. A first maintenance took place in 2015 (EFSA at al., 2016a) followed by another extensive one in 2016-2018. The updated catalogue (MTX 10.0) and the technical report that summarized all changes performed during the maintenance in 2016-2018 were published (EFSA, 2019b).

Moreover, harmonized data collection in the areas of pesticides, contaminant occurrence, veterinary medicinal product residues and food additives (EFSA, 2020) is reflected in the maintenance process of the FoodEx2 catalogue. Considering that all the above data domains use the FoodEx2 catalogue, there was a need to accommodate different requests. Some changes were performed to enable giving a clear guidance (in terms of FoodEx2 codes) to data providers before the data collection launch date, while others were requested from EFSA staff working in different domains, data providers and network members (hereafter stated as users). Moreover, the Food and Agricultural Organisation (FAO) / World Health Organization (WHO) - FAO/WHO GIFT (Global Individual Food consumption data Tool) team[1] and Tufts University[2] (hereafter stated as stakeholders) who are using FoodEx2 to code food consumption and food composition data collected across the world, requested changes that enriched the catalogue with food items consumed outside Europe.

By using the system, both users and stakeholders, provided constructive comments and requests for addition of missing terms and other type of updates in different sections of the catalogue. Establishing a harmonized data collection, in the areas of pesticides, chemical occurrence, veterinary medicinal product residues and additives also needed to be reflected. All these suggestions were considered for the maintenance purposes. A technical group composed of EFSA staff with expertise in different domains carried out the maintenance process evaluating whether and how to implement the received suggestions in the FoodEx2 system.

The present report describes the outcome of the third maintenance process done in 2019 in order to keep track of the changes in the system done within three minor releases published in 2019 (MTX 10.1, MTX 10.2 and MTX 10.3) and the major release published in January 2020 (MTX 11.0). In order to perform a correct coding, with a reasonable level of detail, this report should be read in conjunction with the technical report 'The food classification and description system FoodEx2 (revision 2)' (EFSA, 2015) where the logic and the structure of the system and the rules for the correct use of FoodEx2 are described.

[1] http://www.fao.org/gift-individual-food-consumption
[2] https://www.efsa.europa.eu/sites/default/files/FSNP%20collaboration.pdf

2. Data and Methodologies

2.1. Data

The FoodEx2 catalogue MTX 10.0 - version published in January 2019 was used as a starting point. The suggestions provided during the period between January 2019 and January 2020 by users and stakeholders of the FoodEx2 system were collected through written feedback and used to update the system.

2.2. Methodologies

Each request for addition or changes in the structure was considered and, based on expert judgment, the technical group decided whether a change was needed or whether the request was already covered by the available terms and facet descriptors. All the changes were made following the structure and the logic of the system as described in the report on the development of FoodEx2 revision 2 (EFSA, 2015).

2.2.1. Tools

The tools used for implementing the maintenance were Microsoft Excel® and the FoodEx2 Catalogue Browser[3] (EFSA, 2019a), a user friendly tool developed in Java by EFSA staff (Evidence Management Unit) to help create FoodEx2 codes and to facilitate the management of the MTX catalogue.

2.2.2. Maintenance process

In order to carry out this third maintenance, the maintenance group performed the listed steps:

- Evaluating comments and proposals
- Deciding on the implementation of the proposals, from internal and external users and stakeholders, based on the following options: adding new terms, extending or better specifying the scope of existing terms or identifying the correct way of implementing the suggestion using already existing terms and rules, deprecating terms and set terms as dismissed i.e. not reportable in one hierarchy but still reportable in others.
- Checking for inconsistencies and ambiguities
- Updating the FoodEx2 catalogue
- Summarising the changes for the present technical report.

3. Maintenance output

3.1. Addition of new terms

The terms requested and added in 2019 maintenance were in total 587. Areas that gained significant number of terms were related to the Avian influenza data collection (bird section) which gained 444 new terms (see section 3.6), Feed hierarchy (35 new terms) and facet F23 – Target-consumer (31 new terms) (see section 3.3).

Minor additions of terms were also performed in different hierarchies and facets related to the following domains:

- Botanicals
- Chemical contaminants
- Food consumption
- Pesticides residues
- Veterinary Medicinal Products Residues (VMPR)

[3] https://www.efsa.europa.eu/en/data/data-standardisation

- Zoonoses

Collaboration with the WHO/FAO GIFT team and Tufts University resulted in addition of 23 and 14 new terms respectively, in addition to three overlapping requests from these two stakeholders.

About 20% of all requests for addition of terms were not taken into considerations for the reasons listed below:

- terms were already present in the catalogue under different name or scientific name. In this case the existing term was amended with the common name or synonym of the scientific name
- Foods/feed items could be described using the existing terms (usually upper level term in the relevant food category), adding adequate facets[4] according to the logic of the system (EFSA, 2015).
- the addition of requested terms was postponed to the next release. This was because the requests received was not well defined and therefore further specifications were needed before a term was decided to be added.

A detailed table with all additions performed during 2019, the hierarchies and facets affected and the rationale behind is available in Appendix A.

3.2. Changes in existing terms

Taking into account all suggestions from users and keeping the consistency of the system, amendments to existing terms were applied. Term extended names, common names and/or scientific names were changed for 86 terms. These changes were: 1) editorial, such as singular/plural or correction of misspelling, 2) update of the term extended name, addition of new common names/aliases and removal of existing common names/aliases to ensure more accurate description of the term and 3) update, addition or removal of the scientific name/aliases to better specify a plant or animal species. All changes are presented in in Table 1.

Table 1: Changes performed on FoodEx2 terms' extended name, common name or scientific name/alias in the 2019 maintenance in order to better specify the scope of the term or to clarify its meaning

Code	New term extended name	Old term extended name
A005G	Rye bread and rolls, refined flour	Rye bread, refined flour
A005H	Rye bread and rolls, wholemeal	Rye bread, wholemeal
A005N	Rye-wheat bread and rolls, refined flour	Rye-wheat bread, refined flour
A005P	Rye-wheat bread and rolls, wholemeal	Rye-wheat bread, wholemeal
A00DJ	Rolled oats, instant	Oat rolled grains, instant
A046A	Chili pickle	Chilli pickle
A058N	Greylag goose (as animal)	Wild goose (as animal)
A05CS	Aubergine (as plant)	Egg plant (as plant)
A061D	Hemp seeds (feed)	Hempseeds (feed)
A061V	Acerolas (as plant)	Barbados cherry (as plant)
A06MD	Chili flavour	Chilli flavour
A06VC	Sweet chili flavour	Sweet chilli flavour
A0BCD	Oat feed (feed)	Oat (feed)
A0BCX	Rice grain (feed)	Brewers' rice (feed)
A0BD4	Rye feed (feed)	Rye (feed)

[4] Upper level terms in the relevant food category can be further described with the use of different facets according to the type of the term: RPC terms using F01 - Source, RPC derivatives using F27 – Source-commodity and Composite foods using F04 – Ingredient.

Code	New term extended name	Old term extended name
A0BDE	Wheat feed (feed)	Wheat (feed)
A0BGA	Carob seed (feed)	Locust bean (seed) (feed)
A0C6Y	Conventional (non-organic) production	Conventional non-intensive production
A0CNE	European turtle dove (as animal)	Ringed turtle dove (as animal)
A151Z	Lobelia nicotianifolia Roth ex Schult. (as plant)	Lobelia nicotianaefolia Heyne (as plant)
A15YT	Thymus saturejoides Coss. (as plant)	Thymus satureioides Coss. (as plant)
A16EF	Whooper swans (as animal)	Swans (as animal)
A00DJ	Instant oats	-
A00JB	Cherry pepper$Cluster pepper$Cone pepper$Chilli peppers	Cherry pepper$Cluster pepper$Cone pepper
A014G	-	Sapucaia nut
A01VA	Swine fat tissue$Pork fat tissue	Swine fat tissue
A046A	Chilli pickle	-
A05CS	Eggplant	-
A05TN	Chilli pepper	-
A061V	Barbados cherry	-
A06MD	Chilli flavour	-
A06VC	Sweet chilli flavour	-
A0B6F	Gutweed	-
A0BCX	Brewer's rice (feed)	-
A0CDT	European starling	European starling$Common starling
A0CHX	Black crowned night heron	-
A0DYB	Oak nut$Oaknut	-
A0EGB	Grain flour	-
A13GZ	Baels	-
A13VL	African fan palm$Deleb palm	-
A15JX	Biriba$Wild sugar apple	-
A171A	Tipo$Tipollo$Poleo$Muña	Tipo$Tipollo$Poleo

$ - separator between two common names

The type of term[5] was changed for five codes to have them aligned with the logic of the system (Table 2). In addition, the scope notes of 64 existing terms were updated for a better and more precise description. These changes mainly reflect updates performed in scientific names found in web pages relevant to the term. Changes on the scope notes are not presented in this report, but they were implemented; therefore; they are included in the FoodEx2 database and appear in the file 'MTX_11.0.xlsx' spreadsheet 'term' containing the complete FoodEx2 catalogue in Appendix B.

Table 2: Changes in types of terms in FoodEx2 catalogue performed during the maintenance in 2019

Code	Term Extended name	Old term type	New term type
A03QY	Simple cereals which have to be reconstituted with milk or other appropriate nutritious liquids	RPC Derivatives/ ingredients	Composite food simple
A0BCX	Brewers' rice (feed)	RPC Derivatives/ ingredients	Raw Primary Commodities (RPC)
A0BD5	Sorghum white (feed)	RPC Derivatives/ ingredients	Raw Primary Commodities (RPC)

[5] The type of the term can be one of the following: Natural sources, Raw Primary Commodities (RPC), RPC Derivatives, ingredients, Composite food simple, Composite food aggregated, Broad or mixed, Facets

Code	Term Extended name	Old term type	New term type
A0CTT	Furred wild game (as part-nature)	Raw Primary Commodities (RPC)	Facets
A0CTV	Feathered wild game (as part-nature)	Raw Primary Commodities (RPC)	Facets

3.3. Amendments in Feed area

3.3.1. Updates on Feed section of the Reporting hierarchy

The Feed section (A0BB9 – Feed) of the Reporting hierarchy was reviewed and updated according to the Commission Regulation (EU) 2017/1017 that amended Regulation (EU) No 68/2013 on the Catalogue of feed materials[6]. This review and update of the Feed section were done within an EFSA Working Group (WG) and it is described in the scientific report Animal dietary exposure: overview of current approaches used at EFSA (EFSA, 2019c). As a result of this review, 35 new terms were added to the Feed hierarchy, listed in Table 3. Moreover, the scope notes, term names, common names and/or scientific names were updated, where needed. In order to keep track of the assigned number of feed terms in the Catalogue of Feed Materials, a new implicit attribute - EUFeedReg was created and added to all 653 terms that originate from this source.

Table 3: New terms added to the feed section of the Reporting hierarchy in accordance with Commission Regulation (EU) 2017/1017

Code	Term extended name
A185X	Maize grits (feed)
A185Y	Fermented soya (bean) protein (concentrate) (feed)
A185Z	High-protein low-cellulose fraction of sunflower meal (feed)
A186A	High-cellulose fraction of sunflower meal (feed)
A186B	Vegetable oils, used by food factory (feed)
A186C	Carob seed husk (feed)
A186D	(Sugar) beet molasses, betaine rich, liquid/dried (feed)
A186E	Rapeseed straw (feed)
A186F	Yucca Schidigera juice (feed)
A186G	Waxy-leaf nightshade meal (feed)
A186H	Powdercellulose (feed)
A186J	Terrestrial invertebrates, dead as feed (feed)
A186K	Starfish meal [sea star meal] (feed)
A186L	Calcium carbonate-magnesium oxide (feed)
A186M	Fish oil stearine [Winterized fish oil] (feed)
A186N	Magnesium gluconate (feed)
A186P	Magnesium pidolate (feed)
A186Q	Potassium pidolate (feed)
A186R	Flint [gizzard] grit (feed)
A186S	Redstone [Gizzard] (feed)
A186T	Product from Lactobacillus species rich in protein (feed)
A186V	Product from Trichoderma viride rich in protein (feed)
A186X	Product from Bacillus subtilis rich in protein (feed)
A186Y	Product from Aspergillus oryzae rich in protein

[6] https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32017R1017&from=IT

Code	Term extended name
A186Z	Polyhydroxybutyrate from fermentation with Ralstonia eutropha (feed)
A187A	Sweet flavoured drink (feed)
A187B	Fruit syrup (feed)
A187C	Sweet flavoured syrup (feed)
A187D	Xylo-oligosaccharide (feed)
A187E	Gluco-oligosaccharide (feed)
A187F	Palmitoylglucosamine (feed)
A187G	Salt of lactylates of fatty acids (feed)
A187H	Hyaluronic acid (feed)
A187J	Gluconic acid (feed)
A18AV	Yeasts products (feed)

Another amendment in the Feed section, done within the same WG, was the mapping of the FoodEx2 terms with feedstuffs included in the Organisation for Economic Co-operation and Development Guidance Document (OECD GD) on overview of residue chemistry studies (OECD, 2009) and on residues in livestock (OECD, 2013), described in the same report (EFSA, 2019c). To accommodate this mapping, a new implicit attribute was created: IFN code. IFN code stands for International Feed Nomenclature and is used for identification of the feed items in the OECD GD. It was assigned to all 45 terms where a direct mapping was found (Table 4).

Table 4: Feed terms directly mapped to the feedstuffs from the OECD GD assigned with IFN codes

Code	Term extended name	IFN code
A05CR	Barley grain (feed)	4-00-549
A05DZ	Common millet (feed)	4-03-102
A05FJ	Oat grain (feed)	4-03-309
A05VX	Wheat grains (feed)	4-05-211
A05XB	Triticale grain (feed)	4-20-362
A05XX	Sorghum; [Milo] (feed)	4-04-383
A07XG	Maize grain (feed)	4-20-698
A07XH	Rye grain (feed)	4-04-047
A0BBV	Maize bran (feed)	5-28-235
A0BC0	Maize gluten (feed)	5-28-242
A0BC1	Maize gluten feed (feed)	5-28-243
A0BCP	Rice bran (feed)	4-03-928
A0BDL	Wheat gluten feed (feed)	5-05-221
A0BE4	Distillers' dried grains (feed)	5-00-518
A0BE6	Brewers' grains (feed)	5-00-516
A0BEP	Cotton seed meal, partially decorticated (feed)	5-01-617
A0BEZ	Linseed meal (feed)	5-02-043
A0BF5	Palm kernel meal (feed)	5-03-486
A0BF8	Rape seed meal (feed)	5-08-136
A0BFB	Safflower seed meal, partially decorticated (feed)	5-26-095
A0BFJ	Soya (bean) meal (feed)	5-20-638
A0BFL	Soya (bean) hulls (feed)	1-04-560
A0BFT	Sunflower seed meal (feed)	5-26-098

Code	Term extended name	IFN code
A0BH8	Sugar beet tops and tails (feed)	2-00-649
A0BHF	Dried (sugar) beet pulp (feed)	4-29-307
A0BJD	Potato pulp, dried (feed)	4-03-775
A0BJR	Apple pulp, pressed; [Apple pomace, pressed] (feed)	4-00-419
A0BK3	Citrus pulp, dried (feed)	4-01-237
A0BL0	Clover meal (feed)	1-01-415
A0BL2	Grass, field dried, [Hay] (feed)	1-02-250
A0BL4	Grass, herbs, legume plants, [green forage] (feed)	2-02-260
A0BL7	Horse bean straw (feed)	2-14-388
A0BL9	Lucerne; [Alfalfa] (feed)	2-00-196
A0BLA	Lucerne field dried; [Alfalfa field dried] (feed)	1-00-054
A0BLD	Lucerne meal; [Alfalfa meal] (feed)	1-00-023
A0BLH	Maize silage (feed)	3-28-345
A0BLJ	Pea Straw (feed)	3-03-596
A0BLX	(Sugar) cane molasses (feed)	4-13-251
A0BS3	Products from the potato processing industry (feed)	4-03-777
A0D7H	Peas (dry seeds) (feed)	5-03-600
A0DBB	Carrot (feed)	2-01-146
A0DLH	Manioc; [tapioca]; [cassava] (feed)	2-01-156
A0DLS	Potatoes (feed)	4-03-787
A0EFK	Sugar beet molasses (feed)	4-30-289
A0EHX	Soya (beans) (feed)	5-64-610

An additional enhancement of the Feed section was the implementation of implicit facets F01 - Source and F27 - Source-commodity. Apart from the F02 – Part-nature and F23 – Target-consumer facets, that were before assigned implicitly to all feed terms, these two additional facets were added as implicit facets to all terms, where needed. The F01 – Source facet was added as an implicit facet to 72 Raw Primary Commodities (RPC). Moreover, these 72 terms have been added (made reportable) in the F27 – Source-commodities facet in order to be assigned as implicit Source-commodities to their derivatives. Therefore, F27 – Source-commodities facet was assigned to 237 RPC derivatives (231 existing and 6 new terms) in the Feed section. These changes are not presented in this report, but they are implemented and therefore present in the MTX 11.0 catalogue attached to this report as Appendix B.

3.3.2. Amendment of the facet F23 – Target-consumer

In order to accommodate data collection on feed and water samples within the VMPR domain, the facet F23 – Target-consumer was expanded. For this work, EFSA staff with expertise in the feed consumption/animal nutrition domain joined the EFSA maintenance technical group. Data reported on target consumers in feed area from the previous years were also considered. As a result, the F23 – Target-consumer facet was expanded by 31 terms. Codes added represent feed types consumed by different animal species considering both their life stage and purpose of raising. The current content of the F23 – Target-consumer facet is presented in Table 5, where previously present codes are in bold grey and new codes are in black. Parent terms have centre alignment and children right alignment.

Table 5: The content of the facet F23 - Target-consumer facet

Code	Term extended name	Scope note
A07TV	**Animal feed**	Animal feed
A07TX	**Feed for food animals**	Feed designed for food animals

Code	Term extended name	Scope note
A07TY	Cattle feed	Feed for cattle
A18DN	Calves feed	Feed for calves which are reared for reproduction, veal production or beef production
A18DP	Fattening cattle feed	Feed for fattening cattle, i.e. animals that have completed the weaning period
A18DQ	Dairy/reproductive cow feed	Feed for dairy/reproductive cows, i.e. adult animals in the reproductive stage (males and females)
A18DR	Other bovinae feed	Feed for bovinae other than Bos taurus, including American buffalo, Buffalo, Water buffalo etc.
A07TZ	Fish feed (food fish)	Feed for fish
A18DS	Salmonids feed	Feed for salmons (marine and fresh water) and rainbow trout
A18DT	Marine finfish feed	Feed for marine finfishes (Osteichthyes other than salmonids)
A18DV	Freshwater fish feed	Feed for freshwater fish other than salmonids
A18DX	Crustaceans feed	Feed for crustaceans, for example shrimps etc.
A07VA	Game feed	Feed for game
A18DY	Game bird feed	Feed for game birds farmed or not
A18DZ	Game mammal feed	Feed for game mammals farmed or not
A07VB	Horse feed	Feed for horses
A07VC	Pig feed	Feed for swine (Suidae/Suina)
A18EA	Suckling piglet feed	Feed for suckling piglets, i.e. young animals
A18EB	Weaned piglet feed	Feed for weaned piglets, i.e. young animals
A18EC	Fattening pig feed	Feed for fattening pigs, i.e. adult animals
A18ED	Reproductive pig feed	Feed for reproductive pigs, i.e. adult animals
A07VD	Poultry feed	Feed for poultry
A18EE	Fattening poultry feed	Feed for fattening poultry, i.e. young animals raised for meat production
A18EF	Young breeder/layer poultry feed	Feed for young breeder/layer poultry
A18EG	Breeding/laying poultry feed	Feed for breeding/laying poultry, i.e. adult animals in the reproductive stage
A18EH	Chicken feed	Feed for chickens
A18EJ	Fattening chicken feed	Feed for fattening chickens, i.e. young chicken raised for meat production
A18EK	Young breeder/layer chick feed	Feed for young breeder/layer chicks
A18EL	Breeding/laying chicken feed	Feed for breeding/laying chickens, i.e. adult birds in the reproductive stage (males or females)
A18EM	Turkey feed	Feed for turkeys
A18EN	Fattening turkey feed	Feed for fattening turkey, i.e. young turkey raised for meat production
A18EP	Young breeder turkey feed	Feed for young breeder turkey

Code	Term extended name	Scope note
A18EQ	Breeding turkey feed	Feed for breeding turkey, i.e. adult birds in the reproductive stage (males or females)
A07VE	**Rabbit feed**	Feed for rabbit
A18ER	Fattening rabbit feed	Feed for fattening rabbits, young animals
A18ES	Reproductive rabbit feed	Feed for reproductive rabbits, adult animals at the reproductive stage
A07VF	**Sheep and goat feed**	Feed for ovine and caprine
A18ET	Lambs reared for reproduction or meat production feed	Feed for lambs reared for reproduction or meat production, i.e. young animals reared either for reproduction/dairy or meat production/fattening
A18EV	Dairy/reproductive sheep feed	Feed for dairy/reproductive sheep, i.e. adult animals used for reproduction and/or milk production
A18EX	Kids reared for reproduction or meat production feed	Feed for kids reared for reproduction or meat production, i.e. young animals reared either for reproduction/dairy or meat production/fattening
A18EY	Dairy/reproductive goat feed	Feed for dairy/reproductive goats, i.e. adult animals used for reproduction and/or milk production
A07VG	**Feed for laboratory animals**	Feed for laboratory animals
A07VH	**Feed for non-food producing animals**	Feed for non-food animals
A07VJ	**Bird feed**	Feed for pet birds
A07VK	**Cat feed**	Feed for cats
A07VL	**Dog feed**	Feed for Dogs
A07VM	**Pet fish feed**	Feed for pet fishes
A07VN	**Other pet feed**	Feed for other pet animals

Extending the target consumer facet enabled more precise description of the terms in the section A0BT0 - Compound feed (feed) of the Reporting hierarchy. In previous versions of the catalogue, all terms under this section (and under the whole Feed hierarchy) had assigned as implicit Target consumer facet the A07TV - Animal feed. In version 11.0 this was amended, and 76 terms under Compound feed – A0BT0 (divided to complete and complementary feed sections) have updated implicit F23 - Target-consumer facet based on the name of the complete/complementary feed (term name). For example, the term A0BT2 - Calves (pre-ruminant) / Complete feed (feed) has now the term A18DN – Calves feed as implicit Target-consumer. If the data provider has more detailed information and can therefore select a more specific Target-consumer facet (child term), the implicit facet will be replaced with the more detailed selected one.

3.4. Amendments to implicit facets

Changes to implicit facets were performed on 417 existing terms. 379 of them are related to facets F01 – Source, F27 – Source-commodities and F23 – Target-consumer (as described in section 3.3). Changes performed on the other 38 terms are presented in Table 6. These are related to the cheese section and ambiguities related to the types of cheese (soft, semi-soft, hard etc.). 12 terms under Soft drinks – A03DZ were assigned implicit facet F04 - Ingredient for better specification and more precise determination of the term scope. The term A0F6Q was updated i.e. previously assigned implicit facet F27.A001D - Rice grain is now replaced with F27. A001C – Rice and similar. This will enable users to be more specific when choosing the type of rice used to produce semolina. The term A0FDH – Quince cheese is updated with more specific implicit facet F04 – Ingredient, i.e. previously assigned F04.A04RK – Fruit used as fruit is replaced with F04.A0DXC – Quinces and similar for more precise determination of the term scope

The implicit facet related to the production method F21.A07RY - Wild or gathered or hunted is removed from the term A01TF – Wild goose fresh meat. Fresh meat of the game goose (that is not wild by production method) should also be described with this term; therefore, this facet should be added explicitly only to the animals that are wild or gathered or hunted as stated in the Chemical monitoring reporting guidance: 2020 data collection (EFSA, 2020).

Table 6: Changes on implicit facets performed on existing terms

Code	Term extended name	Performed change
A02ST	Firm - ripened cheeses	The implicit facet F10.A0CJP - Qualitative-info = Hard is removed
A030C	Cheese, bleu d'auvergne	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A030D	Cheese, bleu de gex	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A030H	Cheese, gorgonzola	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A0CJQ - Qualitative-info = Soft
A030K	Cheese, roquefort	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A030L	Cheese, shropshire blue	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A030M	Cheese, stilton	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A030N	Cheese, valdeon	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A030Q	Cheese, amarelo	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A030R	Cheese, ardrahan	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A030S	Cheese, buche de chevre	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A0CJQ - Qualitative-info = Soft
A030T	Cheese, gubbeen	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A030V	Cheese, livarot	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A0CJQ - Qualitative-info = Soft
A030X	Cheese, pont l'eveque	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A0CJQ - Qualitative-info = Soft
A030Y	Cheese, reblochon	The implicit facet F10.A0CJP - Qualitative-info = Hard is replaced with F10.A169Q - Qualitative-info = Semi-soft
A03EA	Soft drink, with fruit juice (fruit content below the minimum for nectars)	The implicit facet F04.A04RK - Ingredient = Fruit used as fruit is added
A03FA	Soft drink, apricot flavour	The implicit facet F04.A06KE - Ingredient = Apricot flavour is added
A03FB	Soft drink, banana flavour	The implicit facet F04.A06KL - Ingredient = Banana flavour is added
A03FC	Soft drink, cherry flavour	The implicit facet F04.A06MB - Ingredient = Cherry flavour is added
A03FE	Soft drink, grapefruit flavour	The implicit facet F04.A06PC - Ingredient = Grapefruit flavour is added
A03FG	Soft drink, lime flavour	The implicit facet F04.A06QH - Ingredient = Lime flavour is added
A03FJ	Soft drink, orange flavour	The implicit facet F04.A06RR - Ingredient = Orange flavour is added
A03FK	Soft drink, pear flavour	The implicit facet F04.A06SE - Ingredient = Pear flavour is added
A03FL	Soft drink, pineapple flavour	The implicit facet F04.A06SJ - Ingredient = Pineapple flavour is added

Code	Term extended name	Performed change
A03FM	Soft drink, raspberry flavour	The implicit facet F04.A06SV - Ingredient = Raspberry flavour is added
A03FN	Soft drink, mango flavour	The implicit facet F04.A06QN - Ingredient = Mango flavour is added
A03FV	Diet soft drinks with fruit juice	The implicit facet F04.A04RK - Ingredient = Fruit used as fruit is added
A0F6Q	Rice semolina	The implicit facet F27.A001D - Source-commodities = Rice grain is replaced with F27.A001C - Source-commodities = Rice and similar-
A0FDH	Quince cheese	The implicit facet F04.A04RK - Ingredient = Fruit used as fruit is replaced with F04.A0DXC - Ingredient = Quinces and similar-
A01TF	Wild goose fresh meat	The implicit facet F21.A07RY - Production-method = Wild or gathered or hunted is removed

3.5. Updates in the hierarchy of botanicals of FoodEx2

The botanical hierarchy was extensively updated during the second maintenance of the FoodEx2 catalogue in 2016 – 2018, according to the needs of EFSA's Compendium of botanicals database[7] (EFSA, 2019b). In 2019, it was further amended with few minor changes: six synonyms terms were added, editorial changes were performed in five terms (typo errors etc.), two terms were deprecated due to duplication (see section 3.8), one term, previously present only in the MTX hierarchy, was added (made reportable) in the Botanical hierarchy and 14 new terms were added under eight different families: Cactaceae, Apocynaceae, Rosaceae, Rubiaceae, Asparagaceae, Rutaceae, Malvaceae, Malpighiaceae (see Appendix A for the detailed list of new terms). All the above-mentioned changes are implemented and present in the MTX 11.0 catalogue in Appendix B of this report.

3.6. FoodEx2 update in accordance with Avian Influenza Surveillance Programmes

FoodEx2 is used for reporting data under the Avian Influenza (AI) surveillance programmes in poultry and wild birds carried out by Member States according to the Commission Decision 2010/367/EU[8] and guidelines provided by Council Directive 2005/94/EC[9] (EFSA, 2018). During 2019 an extensive amendment has been performed in the Bird section – A057X – Bird (as animal). Data provider's requests were collected and evaluated and, as a result, 444 new terms of different bird species were added under the hierarchy term Bird (as animal) - A057X in the MTX (FoodEx2 Matrix) hierarchy and in the F01 - Source facet detailed in Appendix A. Moreover, this section was revised regarding the taxonomy, and the order of the terms was updated accordingly. Finally, to accommodate more detailed description of terms reported under the AI data collection two new terms were added to the facet F29 - Purpose-of-raising: A17LR – Growers and A17LZ – Game purpose and one new term under facet F02 – Part-nature: A17LQ - Combined swabs (as part-nature) (see Appendix A).

All newly added terms were complemented with their relevant Euring code[10]. Euring is the coordinating organisation for European bird ringing schemes which maintains a databank containing ringing recovery information gathered throughout Europe.

3.7. Update on the facet F21 – Production-method

The facet production method is an important piece of information reported within the pesticide data collection. A clear distinction between the conventional/non-organic production, organic production, intensive production and extensive production was needed. The term extended name of the code A0C6Y was Conventional non-intensive production, which was confusing for data providers who could not

[7] https://www.efsa.europa.eu/en/data/compendium-botanicals
[8] Official Journal of the European Union, Commission Decision of 25 June 2010 on the implementation by Member States of surveillance programmes for avian influenza in poultry and wild birds (notified under document C(2010) 4190) (Text with EEA relevance) (2010/367/EU)
[9] COUNCIL DIRECTIVE 2005/94/EC of 20 December 2005 on Community measures for the control of avian influenza and repealing Directive 92/40/EEC
[10] https://blx1.bto.org/euringcodes/species.jsp

understand whether it is related to conventional in terms of non-organic or non-intensive in terms of extensive. To solve this problem, the name of the term A0C6Y – Conventional non intensive production is changed to Conventional (non-organic) production (Table 1) and the scope note was changed to "Non-organic production conditions including the use of synthetic chemical fertilizers, pesticides, herbicides and other continual inputs and processes". In addition, a new term A18FG - Extensive (non-intensive) production is added to the facet F21 – Production method (Appendix A) accompanied by the scope note "Production conditions following procedures that do not stress the maximisation of the output".

3.8. Terms that were deprecated or dismissed

During the maintenance of the FoodEx2 catalogue in 2019, two terms related to the tea flavours found to be duplicates. As they were reportable in different hierarchies and facets and possibly previously used, they were not deprecated but only dismissed (Table 7). For future reporting of Black tea flavour the term A06KV - Black tea flavour and of Green tea flavour the term A06PD - Green tea flavour should be used. A17AX - Chamaemelum mixtum (L.) All. (as plant) and A17Fp - Moringa peregrina (Forssk.) Fiori (as plant), both under Botanicals, were identified as synonyms of the already existing terms: A17AV - Cladanthus mixtus (L.) Chevall. (as plant)' and A17FQ - Moringa aptera Gaertn. (as plant) respectively and were therefore deprecated. Term A0C14 - FA-17.3 Food supplements supplied in a syrup-type or chewable form from the section A163R – Food additive classes (1333/2008) - facet F33 – Legislative-classes was deprecated, as this category was dropped from the Commission Regulation (EU) 2018/1497 amending Annex II to Regulation (EC) No 1333/2008[11].

Table 7: Terms that were deprecated or dismissed during the 2019 maintenance

Code	Term Extended name
A17AX	Chamaemelum mixtum (L.) All. (as plant) [DEPRECATED]
A17FP	Moringa peregrina (Forssk.) Fiori (as plant) [DEPRECATED]
A0C14	FA-17.3 Food supplements supplied in a syrup-type or chewable form [DEPRECATED]
A06VG	Tea black flavour [DISMISSED]
A06VH	Tea green flavour [DISMISSED]

3.9. Revision of applicability or position of existing terms in specific hierarchies and/or facets

In order to adapt the catalogue according to the user's and stakeholder's needs some of existing terms were added to other hierarchies and/or facets. One of the existing terms, previously present only in the MTX hierarchy, was added to the Botanicals hierarchy (section 3.5). Additionally, two terms previously reportable only in the MTX hierarchy were made reportable in the F02 - Part-nature facet.

Due to revision of the position for some existing terms, parent code was changed for some existing terms in one or more hierarchies and/or facets. For example, as described in section 3.6, terms in bird section are reordered according to taxonomy. These minor changes are not presented in this report, but they were implemented; therefore, they are included in the FoodEx2 catalogue and appear in Appendix B file 'MTX_11.0.xlsx' spreadsheet 'term' that contains the complete FoodEx2catalogue.

4. Conclusions

In 2019 the FoodEx2 catalogue was updated following requests from users and needs of the following domains: Pesticides residues, VMPR, Food consumption, Chemical contaminants, Zoonoses and Botanicals.

Based on suggestions provided by users and stakeholders and requests for addition of missing terms, the following modifications were included in the 2019 FoodEx2 maintenance:

[11] https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32018R1497&from=IT

- New terms (587) were added in different hierarchies and facets out of which 444 related to bird species under the AI surveillance programme
- Term names, common names, scientific names and scope notes of some existing terms were changed to better specify the scope of the term or to expand its meaning
- Implicit facets were amended/updated for 417 terms
- A major revision was undertaken in the Feed hierarchy and the facet F23 – Target-consumer, and within the bird section.
- Minor changes were performed in the Botanical hierarchy as well.
- A minor amendment of facet F21 – Production-method was performed to enable clear distinction between Organic vs Conventional (non-organic) and Intensive vs Extensive (non-intensive) production.
- Two duplicates and one unused term were deprecated
- Two additional implicit attributes were added to the FoodEx2 catalogue in order to accomodate the mapping with OECD GD classification of feedstuff (IFN code) and to keep the track of the number assigned to each feed item from the Catalogue of feed materials (EUFeedReg).
- The use of two terms was changed into dismissed or not reportable in MTX, Reporting, Zoonoses and Exposure hierarchy and in the Ingredient facet due to duplication.
- The applicability of some terms was extended to further hierarchies and/or facets, e.g.
- The position of some terms in the hierarchy tree was changed since a better parent-child term relationship was found

The present report documents the update of FoodEx2 performed within three minor releases published in 2019 (MTX 10.1, MTX 10.2 and MTX 10.3) and the major release published in January 2020 (MTX 11.0).

The complete content of the FoodEx2 system is available in Appendix B to this document as exported spreadsheet MTX_11.0.xlsx[12].

[12] The MTX_11_0.ecf is available at https://zenodo.org/record/3243215#.Xj1dNmhKjD4 in the DCF_catalogues.zip folder. Name of the file is MTX.xlsx

References

EFSA, 2011a. Report on the development of a food classification and description system for exposure assessment and guidance on its implementation and use. EFSA Journal 2011;9(12):2489, 84 pp., doi:10.2903/j.efsa.2011.2489

EFSA, 2011b. The food classification and description system FoodEx 2 (draft revision 1). Supporting Publications 2011:215, 438 pp. Available online: http://www.efsa.europa.eu/en/supporting/doc/215e.pdf

EFSA, 2015. The food classification and description system FoodEx 2 (revision 2). EFSA supporting publication 2015: EN-804. EFSA Supporting Publications, 2015: EN-804. 90. https://doi.org/10.2903/sp.efsa.2015.EN-804

EFSA, Vernazza F and Magliano I, 2016a. FoodEx2 annual maintenance 2015. EFSA supporting publication 2016:EN-1049, 24 pp.

EFSA (European Food Safety Authority), Brouwer A, Huneau A, Kuiken T,Staubach C, Stegeman A, Baldinelli F, Verdonck, F and Aznar I, 2018. Scientific report: Reporting Avian Influenza surveillance. EFSA Journal 2018;18(11):5493, 26 pp. https://doi.org/10.2903/j.efsa.2018.5493ISSN:1831-4732

EFSA (European Food Safety Authority), Ioannidou S,2019a. EFSA Catalogue Browser User Guide. EFSA supporting publication 2019:EN-1726. 39pp. doi:10.2903/sp.efsa.2019.EN-1726

EFSA (European Food Safety Authority); Ioannidou S., Nikolic M., 2019b. FoodEx2 maintenance 2016-2018. EFSA supporting publication 2019:EN-1584. 23 pp.

EFSA (European Food Safety Authority), Ardizzone M, Binaglia M, Cottrill B,Cugier J-P, Ferreira L, Gomez Ruiz JA, Innocenti M, Ioannidou S, Lopez Puente S, Merten C, Nikolic Mand Savoini G, 2019c. Scientific report on the animal dietary exposure: overview of current approaches used at EFSA. EFSA Journal 2019;17(11):5896, 18 pp. https://doi.org/10.2903/j.efsa.2019.5896

EFSA (European Food Safety Authority), 2020. Chemical monitoring reporting guidance: 2020 data collection. EFSA Supporting publication 2020.EN-1796. 101 pp. doi:10.2903/sp.efsa.2020.EN-1796

OECD (Organisation for Economic Co-operation and Development), 2009. Guidance document on overview of residue chemistry studies (as revised in 2009). Series on Testing and Assessment number 64 and Series on Pesticides number 32, OECD Environment, Health and Safety Publications, Paris, ENV/JM/MONO (2009) 31, 93 pp.

OECD, 2013. Guidance document on residues in livestock. In: Series on Pesticides No 73. ENV/JM/MONO(2013)8, 4 September 2013.

Abbreviations

AI	Avian Influenza
EC	European Commission
EFSA	European Food Safety Authority
EU	European Union
FAO	Food and Agriculture Organisation of the United Nations
FoodEx2	Food Classification and Description System
IFN	International Feed Nomenclature
MTX	FoodEx2 Matrix hierarchy
OECD	Organisation for Economic Co-operation and Development
RPC	Raw primary commodity
VMPR	Veterinary Medicine Product Residues
WG	Working Group
WHO	World Health Organization

Appendix A – New terms added during the 2019 maintenance

Appendix B – FoodEx2 catalogue MTX_11.0